# Chartly

## Business Requirements Document

*BRD*

| Field | Details |
|---|---|
| **Project Name** | Chartly |
| **Version** | 1.1 (Revised Draft) |
| **Date** | August 8, 2026 |
| **Project Status** | Planning Phase |
| **Platform Type** | Internal Enterprise Tool |

# 1. Project Overview

Chartly is an internal data visualization platform designed to help analysts move from raw datasets to presentation-ready charts through a guided analytical workflow. Users upload CSV or Excel files, review the structure of the data, select the chart type and the columns to be analyzed, configure analysis options, and then generate a chart.

Chartly supports chart regeneration, version history, export, reusable templates, a shared template library, and role-based access control. The product aims to reduce repetitive work in chart creation while improving consistency, traceability, and reusability.

# 2. Project Objectives

- Accelerate the process of creating tables and data charts for data analysts.
- Reduce reliance on scattered external tools for creating charts and reports.
- Build a shared, reusable template (Templates) library at the team level.
- Ensure visual consistency (Branding) through the ability to customize colors and titles on templates.
- Provide a complete audit trail for chart generation and regeneration operations.
- Manage platform access through clear roles and permissions (Admin / Analyst / Viewer).

# 3. Project Scope

## In Scope — Phase 1 and Beyond

- Uploading data files (e.g., CSV, Excel).
- Entering Guidance to clarify the objective of each analysis or chart.
- Support for standard chart types (bar, line, pie).
- Regenerating charts, while retaining a complete history of all previous attempts.
- Exporting final charts as images (PNG/JPG).
- Saving any chart as a template (code) in a shared team/company library.
- Customizing saved templates: colors, title, and other details.
- Managing user roles and permissions (Admin / Analyst / Viewer).

## Out of Scope — Phase 1

- Export as PDF, PPTX, or Excel files.
- Hosting an internal (on-premise) AI engine.
- Sharing templates outside the company's boundaries (a public template library/marketplace).

## 4. Users & Roles (User Roles & Permissions)

| Role | Description | Key Permissions |
|---|---|---|
| **Admin** | Manages the platform, users, and general settings. | Manages users and roles, manages platform settings and connected data sources, in addition to all Analyst permissions. |
| **Analyst** | Creates and manages analyses and charts. | Uploads data, defines Guidance, generates and regenerates charts, saves and edits templates. |
| **Viewer** | Views final outputs only. | Views charts, reports, and templates created by others only, with no create or edit permissions. |

## 5. Core Workflow

**1. Login:** The user logs into the platform, and their permissions are applied according to their role (Admin/Analyst/Viewer).

**2. Data Input:** The analyst uploads a data file.

**3. Guidance:** The analyst selects the required chart type, then specifies the columns and fields to be analyzed, and configures the appropriate analysis options such as aggregation type, sorting, and time granularity as needed. The system then validates the settings and automatically generates the chart.

**4. Review:** The analyst evaluates the output and decides whether it meets the requirement.

**5. Regeneration (Optional):** If the result is not suitable, the analyst adjusts the settings and regenerates the chart; every attempt is saved in the audit trail.

**6. Approval & Export:** Once the desired result is achieved, the chart is exported as an image.

**7. Save as Template (Optional):** The analyst saves the final design as template code in the shared team library.

**8. Customization & Sharing:** Colors/title on the template can be edited, and it becomes available to the rest of the team.

## 6. Detailed Functional Requirements (Functional Requirements)

### 6.1 Data Management

| Priority | Requirement | ID |
|---|---|---|
| **High** | The system shall allow the analyst to upload data files in CSV or Excel format for use as an analysis source. | **FR-1.1** |
| **High** | The system shall automatically analyze the data structure after the file is uploaded, including column names, data types, and missing values where applicable. | **FR-1.2** |

| Priority | Requirement | ID |
|---|---|---|
| High | The system shall allow the analyst to preview a sample of the data after upload and before starting the analysis. | FR-1.3 |
| Medium | The system shall allow the analyst to manually edit a column's data type in cases where automatic type detection is incorrect. | FR-1.4 |

### 6.2 Guided Analysis Builder

| Priority | Requirement | ID |
|---|---|---|
| High | The system shall allow the analyst to select the desired chart type from a list of supported chart types. | FR-2.1 |
| High | The system shall display the columns and fields that are compatible with the selected chart type. | FR-2.2 |
| High | The system shall allow the analyst to specify the columns and fields to be analyzed, such as Dimensions and Metrics. | FR-2.3 |
| High | The system shall allow the selection of an appropriate aggregation operation for numeric data, such as Sum, Average, Count, Minimum, and Maximum, as needed. | FR-2.4 |
| Medium | The system shall provide additional options such as sorting and time granularity according to the chart type and data used. | FR-2.5 |
| High | The system shall prevent or disable the selection of fields or settings that are incompatible with the data type or chart type. | FR-2.6 |
| Medium | The system shall provide rule-based recommendations for selecting the appropriate chart type based on the data type and selected fields. | FR-2.7 |
| Medium | The system shall display a summary of the analysis settings before executing the generation process. | FR-2.8 |
| High | The system shall validate the analysis settings before creating the chart. | FR-2.9 |

### 6.3 Chart Generation & Management

| Priority | Requirement | ID |
|---|---|---|
| High | The system shall create the chart based on the chart type, columns, and analysis options specified by the analyst. | FR-3.1 |
| High | The system shall display the chart interactively within the platform. | FR-3.2 |
| High | The system shall allow the analyst to modify analysis settings and regenerate the chart. | FR-3.3 |
| High | The system shall retain chart versions resulting from regeneration operations. | FR-3.4 |

| Priority | Requirement | ID |
|---|---|---|
| High | The system shall log the date and time of each generation or regeneration operation, along with the analysis settings used. | FR-3.5 |
| High | The system shall allow the final chart to be exported in PNG or JPG format. | FR-3.6 |

### 6.4 Templates Library

| Priority | Requirement | ID |
|---|---|---|
| High | The system shall allow the analyst to save the settings of a successful chart as a reusable template. | FR-4.1 |
| High | Saved templates shall be available to authorized team members. | FR-4.2 |
| Medium | The system shall allow editing of a template's title, colors, and basic display settings. | FR-4.3 |
| Medium | The system shall allow searching within the template library by template name or chart type. | FR-4.4 |
| Medium | The system shall verify a template's compatibility with a new dataset before applying it. | FR-4.5 |
| Medium | The system shall allow mapping of template fields to the corresponding columns in a new dataset when column names differ. | FR-4.6 |

### 6.5 User & Access Management

| Priority | Requirement | ID |
|---|---|---|
| High | The system shall allow the system administrator to add users, suspend their accounts, and assign their roles (Admin / Analyst / Viewer). | FR-5.1 |
| High | A Viewer shall be able to view only the charts and templates they are authorized to access, without the ability to edit or regenerate them. | FR-5.2 |
| High | An Analyst shall be able to create, edit, and analyze charts and templates according to the permissions granted to them. | FR-5.3 |
| High | An Admin shall be able to manage users, roles, shared templates, and platform settings. | FR-5.4 |

### 6.6 Audit Trail & Version History

| Priority | Requirement | ID |
|---|---|---|
| High | The system shall log all chart creation and regeneration operations. | FR-6.1 |
| High | The audit trail shall include the username, date, time, and analysis settings used for each operation. | FR-6.2 |

| Priority | Requirement | ID |
|---|---|---|
| Medium | The system shall allow the analyst to browse previous chart versions and revert to a prior version when needed. | FR-6.3 |
| High | Audit trail records shall be protected from modification or deletion by regular users. | FR-6.4 |

# 7. Non-Functional Requirements

## 7.1 Security & Access Control

| Priority | Requirement | ID |
|---|---|---|
| High | Data files and analysis results shall be protected from unauthorized access, with appropriate encryption applied to data in transit and at rest. | NFR-1.1 |
| High | Admin / Analyst / Viewer permissions shall be enforced at the level of screens, functions, and actions. | NFR-1.2 |
| High | Users shall be prevented from performing any operation outside the scope of their granted permissions. | NFR-1.3 |

## 7.2 Data Privacy

| Priority | Requirement | ID |
|---|---|---|
| High | Data analysis and chart generation operations shall be carried out within the environment hosting the platform. | NFR-2.1 |
| High | The data content analysis process shall not rely on sending data to AI services or other external services. | NFR-2.2 |
| High | Appropriate policies shall be applied to protect data files during storage and processing. | NFR-2.3 |

## 7.3 Auditability & Traceability

| Priority | Requirement | ID |
|---|---|---|
| High | Chart creation and regeneration operations shall be logged with the username, date, time, and analysis settings used. | NFR-3.1 |
| High | Audit trail records shall be protected from unauthorized modification or deletion. | NFR-3.2 |
| High | Every chart output shall be traceable to the analysis settings used to create it, including chart type, fields, operations, and settings. | NFR-3.3 |

## 7.4 Performance & Scalability

| Priority | Requirement | ID |
|---|---|---|
| High | Data upload and preview operations shall remain responsive within the data size limits defined for the first release. | NFR-4.1 |
| High | Chart creation and regeneration operations shall remain responsive within the platform's target performance limits. | NFR-4.2 |
| Medium | The platform shall be designed to initially support a single data analysis team, with the ability to scale later to support additional teams and users. | NFR-4.3 |

### 7.5 Reliability & Compatibility

| Priority | Requirement | ID |
|---|---|---|
| High | The platform's core functions, such as data profiling, validation, and chart creation, shall operate without reliance on external services. | NFR-5.1 |
| Medium | The platform shall operate on modern, commonly used browsers, including Chrome, Edge, and Firefox. | NFR-5.2 |
| Medium | Users shall not need to install any additional software on their device to use the platform. | NFR-5.3 |

### 7.6 Usability & Maintainability

| Priority | Requirement | ID |
|---|---|---|
| High | The Guided Analysis Builder shall be clear and easy to use, presenting the analyst only with options relevant to the selected chart type and data. | NFR-6.1 |
| Medium | The validation and recommendation rules engine shall be extensible, allowing new chart types and analysis rules to be added without fundamental changes to the system architecture. | NFR-6.2 |
| High | The platform interface shall be organized so that the user can clearly move from data upload to chart selection, then generation, review, export, and saving as a template. | NFR-6.3 |

## 8. Proposed Implementation Phases

### Phase 1 — MVP

- Uploading files (CSV/Excel) as a data source.
- Defining Guidance and generating standard charts (bar, line, pie).
- Exporting the chart as an image, and regeneration with a basic audit trail.
- Saving templates and sharing them within the team (without advanced customization yet).
- Basic roles and permissions (Admin/Analyst/Viewer).

### Phase 2

- Support for advanced/interactive charts (dashboards, heatmaps).
- Full template customization (colors, titles, and other details).

### Phase 3

- Performance and scalability improvements based on actual usage results.

## 9. High-level Success Criteria

- Reduced time required to go from raw data to a ready chart (from hours to minutes) — proposed target: reduce this time by at least 70% within the first quarter after launch.
- Actual usage rate of the shared template library by team members.
- Reduced reliance on scattered external tools for creating charts and reports.
- Analyst satisfaction with the platform's accuracy and ease of use, with continuous improvement based on periodic feedback.

## 10. Risks & Assumptions

### 10.1 Assumptions

- The users of the first release are a single data analysis team of limited size, allowing the MVP to launch without the need for advanced scalability improvements right away.
- User data will be uploaded in standard CSV/Excel formats (without complexities such as pivot tables or multiple linked sheets) during Phase 1.
- The internal infrastructure required (hosting, storage, network permissions) to run the platform as an internal tool is available, without reliance on external cloud services.

## 11. Dependencies

- Availability of a ready internal hosting environment (On-premise/Internal Infrastructure) before development begins, ensuring compliance with NFR-2.1 and NFR-2.2.
- Assignment of a technical team responsible for designing and maintaining the chart engine and the validation/recommendation rules system referenced in Section 1.

## 12. Stakeholders

| Role | Key Responsibility | Name / Entity |
|---|---|---|
| **Product Owner** | Sets priorities and gives final approval on requirements. | TBD |
| **Tech Lead** | Technical decisions, solution design, and architecture. | TBD |
| **Analyst Representative** | Represents end-user needs and validates usability. | TBD |
| **Security / Compliance Officer** | Reviews security and data privacy requirements (Sections 7.1 and 7.2). | TBD |

## 13. Document Sign-off

| Role | Name | Signature | Date |
|---|---|---|---|
| **Product Owner** | | | |
| **Tech Lead** | | | |